Project Report — Amazon Sales Analysis

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1. Cover Page

Field	Details
Project Title	Amazon Product Sales Analysis
Sector	E-Commerce / Retail Technology
Team ID	DETECTIVES
Team Members	Vipul · Ananya Gupta · Yash Mali · Abhay Pratap Yadav
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Institute	Newton School Of Technology
Submission Date	28 April 2026

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2. Executive Summary

Problem

Amazon sellers and category managers lack a data-driven view of which product attributes (price point, discount depth, rating, reviews, badge status, coupons, and sponsorship) actually drive monthly purchase volume and estimated revenue across product categories.

Approach

We sourced a raw Amazon product dataset (~42,675 listings), performed a full ETL pipeline (extraction → cleaning → feature engineering), conducted exploratory data analysis across 18+ product categories, and ran five inferential statistical tests (Pearson/Spearman correlation, OLS multiple regression, Welch's t-test, one-way ANOVA, and chi-square). Findings were visualised in a Tableau dashboard.

Key Insights

- **Power & Batteries** is the single highest-revenue category, generating an estimated **$526.7M** in the analysis window, driven by very high purchase volumes (~25.9M units/month).
- **91.3% of products** carry a rating ≥ 4.0, and high-rated products account for **95.8% of total estimated revenue** ($1.508B vs $0.067B).
- Only **0.9% of products** hold the "Best Seller" badge, yet they earn significantly higher mean revenue (**$510k vs $48k/month**, Welch's t-test $t = 7.17$, $p = 7.74 \times 10^{-12}$).
- **Discounts have no measurable revenue effect** (Pearson $r = 0.000$, $p = 0.994$) and **coupons are revenue-negative** (OLS –$56,470/month); both budgets should be redirected to sponsored placements.
- Category membership drives revenue more than any single price or discount lever (ANOVA $F = 161.2$, $p < 0.001$).

Key Recommendations

1. **Invest in earning the Best Seller badge** — single largest revenue lever (+$396,500/month OLS; 10.6× mean revenue premium).
2. **Stop using discounts as a revenue lever** — zero correlation with revenue; redirect margin to sponsored placements (+$156,700/month).
3. **Cut coupon budgets and replace with sponsored campaigns** — coupons are revenue-negative (−$56,470/month) and independent of badge status (chi-square p = 0.489).
4. **Concentrate inventory and ad spend on top-revenue categories** — Power & Batteries, Laptops, Smart Home (ANOVA F = 161.2).
5. **Drive review velocity on high-rating, low-review SKUs** — +$1.74/review, near-zero marginal cost.

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3. Sector and Business Context

Sector Overview

E-commerce is the fastest-growing retail channel globally. Amazon dominates with >38% US market share. Products compete on algorithmic discoverability (ranking, badge signals, buy-box ownership) as much as on price. Data-driven sellers who optimise pricing, reviews, and promotional levers outperform reactive ones by significant margins.

Decision-Maker / Stakeholder

- **Primary:** Amazon third-party sellers and brand managers seeking to maximise product-level revenue.
- **Secondary:** Category managers at Amazon who monitor segment health and competitive positioning.
- **Tertiary:** Retail investors tracking category-level demand trends.

Why This Problem Matters

Without a structured, statistical view of what drives revenue, sellers default to intuition — discounting aggressively with unclear ROI, ignoring review strategy, or entering saturated categories. This project quantifies each lever independently so decisions can be evidence-based and ROI-focused.

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4. Problem Statement and Objectives

Formal Problem Definition

> *Which combination of product attributes — price, discount depth, rating, review count, best-seller badge, sponsorship, and coupon presence — most significantly determines estimated monthly revenue for an Amazon product listing, and how does this relationship vary across product categories?*

Scope

- **In scope:** Product listings scraped from Amazon (August–September 2025), electronics-adjacent categories.
- **Out of scope:** Seasonal trend modelling (limited date range), international marketplaces, advertising cost data.

Success Criteria

Criterion	Target
Statistical significance of identified revenue drivers	$p < 0.05$ on ≥ 3 of 5 hypothesis tests
EDA coverage	≥ 10 meaningful visual insights
Dashboard usability	Executive view + Operational drill-down with filters
Data quality	Retain ≥ 65% of raw rows after cleaning

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5. Data Description

Source Citation and Access Link

- **Dataset:** Amazon Product Sales Data (web scrape — single snapshot)
- **Collection date:** 2025-08-21 (see `data_collected_at` column)
- **Raw file:** `data/raw/amazon_products_sales_data_raw.csv`
- **Full column reference:** `docs/data_dictionary.md`

Dataset Size and Coverage

Metric	Value
Raw rows	42,675
Cleaned rows	30,228
Retention rate	70.8%
File size (raw)	~36 MB
File size (cleaned)	~8.2 MB
Categories covered	15+ electronics sub-categories

Key Columns

Column	Type	Source	Description
`product_title`	String	raw	Full product name as shown on Amazon
`product_rating`	Float	raw	Average star rating; observed range 2.0–5.0
`total_reviews`	Float	raw	Total customer reviews on the listing

Column	Type	Source	Description
`purchased_last_month`	Float	raw	Units sold in past 30 days — **never imputed**, rows with null are dropped
`discounted_price`	Float	raw	Current selling price (USD)
`original_price`	Float	raw	Listed MSRP / pre-discount price (USD)
`discount_percentage`	Float	raw	Pre-computed discount %; validated ≤ 90; observed range 0–85.42
`is_best_seller`	String	raw	Badge text, lowercased. Treat `"best seller"` as the true badge flag
`is_sponsored`	String	raw	`"sponsored"` or `"organic"`
`has_coupon`	String	raw	Coupon text, or `"no coupon"`. Anything ≠ `"no coupon"` = has coupon
`buy_box_availability`	String	raw	`"Add to cart"` or `"No"` (nulls filled with `"No"`)
`delivery_date`	String	raw	ISO date string — 17.8% null; re-parse with `pd.to_datetime()`
`sustainability_tags`	String	raw	Sparse (92.1% null); common values: Small Business, Carbon impact
`data_collected_at`	String	raw	Scrape timestamp (`2025-08-21`) — re-parse on load
`product_category`	String	raw	Normalised product category (lowercase + stripped)
`estimated_revenue`	Float	**engineered**	`purchased_last_month × discounted_price` — **primary KPI**
`discount_value`	Float	**engineered**	`original_price - discounted_price` (absolute $ saved)
`rating_category`	String	**engineered**	`"High"` if rating ≥ 4.0, else `"Low"`

Data Quality Issues

- **~12,400 rows dropped:** missing values in core fields (`discounted_price`, `original_price`, `product_rating`, `purchased_last_month`) — net effect 42,675 → 30,228 rows (29% loss, predominantly missing sales counts).
- **Numeric columns stored as strings** (e.g., prices with `$` symbols) — required `pd.to_numeric()` coercion.
- **Boolean-like flags** (`is_best_seller`, `is_sponsored`, `has_coupon`) stored as raw lowercased badge strings — downstream analysis derives 0/1 flags in-notebook without modifying the CSV.
- **Date columns written as strings** after `df.to_csv()` — always call `pd.to_datetime()` on `delivery_date` and `data_collected_at` when loading `cleaned_data.csv`.

- **Discount % is pre-computed** by the scraper and may diverge slightly from `(original - discounted) / original × 100` due to time-of-scrape promotions.
- **Sustainability tags** too sparse (~8% present) for standalone analysis.
- Rows with `discount_percentage > 90` treated as data artefacts and removed.

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6. Cleaning and Transformation

Major Cleaning Steps

Step	Action	Rows Before	Rows After
1	Drop exact duplicates	42,675	~42,500
2	Coerce 6 numeric columns (`pd.to_numeric`)	—	—
3	Parse 2 date columns (`pd.to_datetime`)	—	—
4	Fill `buy_box_availability` NaN → "No"	—	—
5	Drop rows missing core fields	~42,500	~31,000
6	Validate ranges (price > 0, rating 0–5, discount ≤ 90%)	~31,000	30,228
7	Normalise `product_category` (lowercase + strip)	—	—
8	Lowercase boolean-like flag columns	—	—
9	**Engineer** `estimated_revenue`, `discount_value`, `rating_category`	—	—
10	Drop non-analytical columns (`product_image_url`, `product_page_url`)	—	—

Assumptions Made

- `estimated_revenue = purchased_last_month × discounted_price` is a conservative proxy for revenue (ignores returns, fees, and multi-channel sales).
- `product_rating ≥ 4.0` qualifies as "High" quality; below 4.0 is "Low".
- `buy_box_availability` NaN treated as "No" (product does not hold Buy Box by default).
- Rows with `discount_percentage > 90%` are data artefacts, not genuine listings.

Output Dataset Description

- **File:** `data/processed/cleaned_data.csv`
- **Shape:** 30,228 rows × 18 columns
- **Final Tableau export:** `data/processed/final_tableau_data.csv`

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7. KPI Framework

KPI	Formula	Why It Matters
Estimated Revenue (per product)	`purchased_last_month × discounted_price`	Primary demand signal; proxy for seller monthly earnings
Total Category Revenue	`Σ estimated_revenue` per category	Identifies highest-value market segments
Average Discount Depth	`mean(discount_percentage)`	Indicates competitive pricing pressure
Rating Score (avg)	`mean(product_rating)`	Consumer trust and algorithmic ranking factor
Review Volume	`mean(total_reviews)`	Social proof and discoverability metric
Best Seller Penetration	`count(is_best_seller == 'best seller') / total`	Badge capture rate; quality signal
Coupon Adoption Rate	`count(has_coupon != 'no coupon') / total`	Promotional activity level
Buy Box Win Rate	`count(buy_box_availability != 'No') / total`	Competitive advantage on the listing page
Sponsorship Rate	`count(is_sponsored == 'sponsored') / total`	Paid visibility investment
High-Rating Share	`count(rating_category == 'High') / total`	Portfolio quality health

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8. Exploratory Analysis

Major Trends

Observation	Value
Total estimated revenue across all products	**$1.575B**
Average product rating	**4.44** (median: 4.5)
Average discounted price	**$156.62**
Average original price	**$172.90**
Average discount depth	**7.78%**
Average monthly purchases per product	**1,354.57 units**
Products with rating ≥ 4.0 ("High")	**91.3%** (27,580 / 30,228)
Products holding Buy Box	**77.0%**
Sponsored products	**11.9%**

Observation	Value
Products with coupon	**3.7%**
Best Seller badge holders	**0.9%**

Segment-Level Insights

Top 5 Categories by Estimated Revenue:

Rank	Category	Estimated Revenue
1	Power & Batteries	$526.7M
2	Laptops	$312.4M
3	Phones	$240.1M
4	Other Electronics	$141.0M
5	Cameras	$87.0M

Top 5 Categories by Product Count:

Rank	Category	Product Count
1	Laptops	6,038
2	Other Electronics	5,295
3	Phones	5,108
4	Power & Batteries	2,683
5	Cameras	2,490

> Power & Batteries has only the 4th-highest product count but the #1 revenue position — driven by extremely high average purchase volume per product.

High vs Low Rating Revenue Split:

- High-rated products (≥ 4.0): **$1,508.2M (95.8% of total revenue)**
- Low-rated products (< 4.0): **$66.6M (4.2%)**

Visual Summaries

(Refer to Tableau dashboard for interactive equivalents)

- Bar chart: Category revenue comparison
- Histogram: Distribution of product ratings (left-skewed, peaked at 4.5)
- Scatter: `total_reviews` vs `estimated_revenue` (positive correlation)
- Box plot: Revenue by `rating_category`
- Treemap: Revenue share by category

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9. Statistical Analysis

Methods Used

Five hypothesis tests were conducted on `data/processed/cleaned_data.csv` (n = 30,228). Actual statistics are taken from the executed `notebooks/04_statistical_analysis.ipynb`; α = 0.05 throughout.

#	Hypothesis	Test Used
H1	Reviews predict revenue	Pearson + Spearman correlation + OLS coefficient
H2	Discounts drive sales	Pearson + Spearman correlation + OLS coefficient
H3	Best Seller badge → higher revenue	Welch's two-sample t-test
H4	Revenue differs across product categories	One-way ANOVA
H5	Coupons and Best Seller status are not independent	Chi-square test of independence

Results

#	Hypothesis	Statistic	p-value	Decision
H1	Reviews → Revenue	Pearson r = 0.208; OLS coef = **+$1.74/review**	< 0.001 / < 0.001	**Reject H_0** — reviews significantly predict revenue
H2	Discount % → Revenue	Pearson r = **0.000**; OLS coef = −$58.4/pp	0.994 / 0.391	**Fail to reject H_0** — discounts have **no detectable revenue effect**
H3	Best Seller badge premium	t = **7.17**, means $510k vs $48k (**10.6× ratio**)	**7.74×10^{-12}**	**Reject H_0** — best sellers earn significantly more
H4	Category revenue differences	F = **161.2**	< 0.001	**Reject H_0** — categories differ significantly
H5	Coupon ↔ Best Seller independence	χ^2 = **0.480**, dof = 1	**0.489**	**Fail to reject H_0** — coupons and Best Seller status are independent

Additional OLS findings (Model R^2 = 0.180, F = 944.9, p < 0.001):

Predictor	OLS Coefficient	p-value	Interpretation
`flag_best_seller`	**+$396,500/month**	< 0.001	Largest independent effect in the model

Predictor	OLS Coefficient	p-value	Interpretation
`flag_sponsored`	**+$156,700/month**	< 0.001	Strongest controllable positive lever
`flag_has_coupon`	**−$56,470/month**	< 0.001	Coupons are revenue-negative once other factors controlled
`total_reviews`	**+$1.74/review**	< 0.001	Moderate, reliable positive effect
`discount_percentage`	−$58.4/pp	0.391	Not significant — discounting does not drive revenue

Business Interpretation

- **Best Seller badge** is the single largest controllable revenue lever (+$396,500/month, OLS; 10.6× mean revenue premium, t-test). Earning and maintaining the badge should be a top operational priority.
- **Sponsored placement** is the strongest controllable positive lever that does not require badge status (+$156,700/month). Budget should be shifted toward sponsored campaigns.
- **Discounts do not drive revenue** — Pearson r = 0.000, OLS coefficient insignificant (p = 0.391). Discount-led promotion is unsupported by the data.
- **Coupons are revenue-negative** in the regression (−$56,470/month, p < 0.001) and are **independent of Best Seller status** (chi-square p = 0.489) — coupons do not help win the badge and actively erode revenue; coupon budgets should be redirected to sponsored placements.
- **Review count** carries a reliable, low-cost marginal effect (+$1.74/review, p < 0.001); review-velocity campaigns on high-rating, low-review SKUs offer the highest ROI of the continuous levers.
- **Category membership** (ANOVA, F = 161.2, p < 0.001) explains more revenue variance than any individual product attribute — entering the right category is the highest-impact strategic decision.

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10. Dashboard Walkthrough

Dashboard Objective

Provide sellers and category managers with an interactive, decision-ready view of Amazon product performance across all categories, enabling rapid identification of high-revenue segments, quality gaps, and promotional opportunities.

Executive View

- **KPI tiles:** Total estimated revenue, average rating, average discount %, buy-box win rate.
- **Category revenue bar chart:** Ranked by total estimated revenue, colour-coded by listing count.
- **Rating distribution:** Histogram showing overall product quality health.

Operational View

- **Product-level table:** Sortable by revenue, rating, reviews, discount %, with badge filter.

- **Scatter plot:** `total_reviews` vs `estimated_revenue` coloured by `rating_category`.
- **Box plot:** Revenue distribution by `product_category` (outlier visibility).

Filters and Interactivity

- **Category filter:** Select one or multiple product categories.
- **Rating range slider:** Filter by rating range (e.g., 4.0–5.0 only).
- **Badge filter:** Best Seller / No Badge / Limited Time Deal.
- **Price range:** Min–Max discounted price.
- **Buy Box toggle:** "Buy Box available" vs "Not available."
- **Sponsored filter:** Sponsored / Organic.

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11. Key Insights

1. **Power & Batteries is the dominant revenue category** — $526.7M total estimated revenue; mean revenue per listing = **$196,303** (nearly 4× the next category). The high mean is driven by a small number of very high-volume bulk SKUs.
2. **Best Seller badge = 10.6× revenue premium** — badge holders average **$510,036/month** vs $48,141/month for non-badge listings (Welch's t-test, $t = 7.17$, $p = 7.74 \times 10^{-12}$).
3. **Sponsored placement is the strongest controllable lever** — OLS coefficient **+$156,700/month** ($p < 0.001$), the highest controllable positive effect excluding the Best Seller badge itself.
4. **Discounts do NOT drive revenue** — Pearson $r = 0.000$ ($p = 0.994$), OLS coefficient −$58.4/pp ($p = 0.391$). Discount-led promotion is statistically unsupported.
5. **Coupons are revenue-negative** once other factors are controlled — OLS coefficient **−$56,470/month** ($p < 0.001$); chi-square confirms coupons and Best Seller status are **independent** ($\chi^2 = 0.480$, $p = 0.489$), meaning coupons don't help win the badge.
6. **Review count carries a reliable, low-cost marginal effect** — OLS coefficient **+$1.74 per additional review** ($p < 0.001$); second-strongest continuous predictor.
7. **91.3% of products are high-rated (≥ 4.0)** — quality is a baseline entry requirement, not a differentiator; differentiation happens through reviews, sponsorship, and badge status.
8. **High-rated products generate 95.8% of all estimated revenue** ($1.508B of $1.575B) — a rating < 4.0 is a disqualifying condition.
9. **Category membership explains more variance than any single attribute** — ANOVA $F = 161.2$, $p < 0.001$. Top categories by mean revenue: Power & Batteries ($196k), Laptops ($52k), Smart Home ($50k).
10. **Laptops have the most listings (6,038) but only rank 2nd in revenue** — the category is highly fragmented; per-product mean revenue is 3.8× lower than Power & Batteries.
11. **OLS model $R^2 = 0.180$** — the seven predictors (price, discount, rating, reviews, badge, sponsored, coupon) jointly explain 18% of revenue variance; 82% is attributable to unlisted factors (product age, advertising history, seasonal demand).
12. **Median rating of 4.5 indicates a ceiling effect** — at the top of the market, revenue is decided by review volume, sponsored visibility, and badge status — not by star rating alone.

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12. Recommendations

> All five recommendations are grounded in specific test outputs from `notebooks/04_statistical_analysis.ipynb` and documented in full in `docs/recommendations.md`.

R1 — Invest in Earning the Best-Seller Badge

Evidence: Welch's t-test: $t = 7.17$, $p = 7.74 \times 10^{-12}$. Best-sellers earn **$510k vs $48k/month** (10.6× ratio). OLS confirms +$396,500/month independent effect ($p < 0.001$) — the single largest coefficient in the model.
Action: Prioritise inventory, fulfilment SLA, review velocity, and listing-quality investment for the top ~10% of category-leading SKUs to push them into Amazon's Best-Seller badge.
Estimated Impact: If 5 non-badge SKUs cross the threshold at half the observed lift: **≈ $1.15M/month** in incremental revenue.

R2 — Stop Using Discounts as a Revenue Lever

Evidence: Pearson $r = 0.000$ ($p = 0.994$); Spearman $\rho = -0.116$ ($p < 0.001$ — slight *negative* rank association); OLS coef = −$58.4/pp ($p = 0.391$, not significant). Discounts have no detectable positive effect on revenue.
Action: Reduce average discount depth on listings where discounting is the primary lever; reallocate that margin to sponsored placements.
Estimated Impact: Cutting average discount by 5 pp across ~7,500 bottom-quartile SKUs recovers ≈ **$1.4M/month** in margin without measurable sales loss.

R3 — Concentrate Inventory and Ad Spend on High-Revenue Categories

Evidence: One-way ANOVA $F = 161.2$, $p < 0.001$. Top categories by mean monthly revenue: Power & Batteries ($196,303), Laptops ($51,740), Smart Home ($50,273).
Action: Reallocate inventory commitment and sponsored-listing budget away from low-mean-revenue categories (Printers & Scanners, Networking) toward the top 3.
Estimated Impact: Shifting 30% of ad budget from bottom-3 to top-3 categories yields an estimated **15–25% uplift** on the reallocated portion.

R4 — Shift Promotion Budget from Coupons to Sponsored Placements

Evidence: OLS: `flag_sponsored` = **+$156,700/month** ($p < 0.001$); `flag_has_coupon` = **−$56,470/month** ($p < 0.001$). Chi-square confirms coupons and Best Seller status are independent ($\chi^2 = 0.480$, $p = 0.489$) — coupons do not help win the badge.
Action: Identify SKUs running coupons that are not simultaneously sponsored; cut the coupon and replace with an equivalent-cost sponsored campaign over 30 days.
Estimated Impact: On a base of 50 listings, shifting from coupon to sponsored recovers the −$56k drag and captures part of the +$157k uplift — a modeled swing of up to **$10M/month** across the portfolio.

R5 — Drive Review Velocity on High-Rating, Low-Review SKUs

Evidence: Pearson $r = 0.208$ ($p < 0.001$); OLS coef = +$1.74/review ($p < 0.001$) — second-strongest continuous predictor, with near-zero marginal cost.
Action: Filter `cleaned_data.csv` for `product_rating ≥ 4.5 AND total_reviews < 500`; deploy post-purchase email + in-package QR insert review-request flow.

Estimated Impact: Moving 10 SKUs from 200 → 1,000 reviews: **10 × 800 × $1.74 ≈ $13,920/month**, or ~$42k over a 90-day campaign.

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13. Limitations and Next Steps

Data Limitations

- **Single time-point snapshot:** Data collected August–September 2025 only. No seasonality analysis is possible with a single month's window.
- **Revenue is a proxy:** `estimated_revenue = purchased_last_month × discounted_price` ignores returns, Amazon fees, taxes, and multi-channel effects.
- **No cost data:** Without COGS or advertising spend, profit margin analysis is not possible.
- **Category labels are scraped strings:** Some category assignments may be imprecise due to seller-defined categorisation inconsistencies.
- **No seller-level data:** All analysis is at the product level; portfolio-level seller strategy cannot be assessed.

Method Limitations

- OLS regression assumes linearity; revenue distributions are heavily right-skewed. Log-transforming the target variable would improve model fit (current $R^2 = 0.180$ likely underestimates true predictability).
- Correlation does not imply causation — e.g., the review-revenue and sponsored-revenue relationships may be inflated by selection bias (sellers sponsor and solicit reviews for already-performing SKUs).
- Large-sample tests ($n = 30{,}228$): even trivially small effects appear statistically significant. Always assess practical significance alongside p-values.
- The chi-square result (H5 fail-to-reject, $p = 0.489$) means coupon and badge status are independent — the previous report version incorrectly stated they were associated; this has been corrected.

Suggested Future Work

1. **Time-series data collection** (6–12 months) to enable seasonality and trend modelling.
2. **Log-linear regression** on revenue for better distributional fit.
3. **Clustering / segmentation** of products using K-means to identify natural product archetypes.
4. **Profit margin modelling** by integrating advertising cost and estimated COGS data.
5. **Natural Language Processing** on `product_title` and `sustainability_tags` to assess keyword and ESG signal effects on revenue.
6. **Expand to non-electronics categories** (Clothing, Home & Kitchen, Books) for cross-sector comparison.

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14. Contribution Matrix

> All members actively collaborated in discussions, validation of results, and final review of the project.

his project was completed as a collaborative effort, with all team members contributing across different stages of the data pipeline, analysis, and dashboard development.

- **Vipul Yadav** Contributed to data cleaning, preprocessing, feature engineering, and final dashboard publishing on Tableau Public.
- **Ananya Gupta** Worked on dashboard design, KPI creation, and development of visualizations in Tableau.
- **Yashveer Singh** Focused on deriving key insights from the data and formulating business recommendations.
- **Abhay Pratap Yadav** Prepared the final project report and ensured proper documentation of analysis and findings.
- **Yash Kishor Mali** Managed project documentation, including README, project structure, data cleaning and overall repository organization.